

EXPERIMENT 6

Aim

To apply and compare K-Means Clustering and Hierarchical Clustering on the Iris dataset in order to group similar data points into clusters based on their features.

Problem Statement

Clustering is an unsupervised learning technique used to group similar data points without predefined labels.

The problem is to identify natural groupings in the Iris dataset based on flower measurements such as sepal length, sepal width, petal length, and petal width.

The goal of this experiment is to:

- Apply clustering algorithms to segment the dataset
- Analyze how well the clusters represent actual species
- Compare the performance of both methods

Dataset

- **Name:** Iris Dataset
- **Link:** <https://www.kaggle.com/datasets/uciml/iris>

Methodology

1. Data Loading

- Loaded Iris dataset using `sklearn.datasets`

2. Data Preprocessing

- Selected relevant numerical features
- Applied **StandardScaler** for normalization

3. K-Means Clustering

- Used **Elbow Method** to determine optimal number of clusters (K)
- Applied K-Means with **K = 3**
- Assigned cluster labels to each data point

4. Hierarchical Clustering

- Used **Ward linkage method**
- Generated **Dendrogram** to visualize cluster formation
- Selected number of clusters = 3
- Applied Agglomerative Clustering

5. Visualization

- Scatter plots for both clustering methods
- Dendrogram for hierarchical clustering

Result

Performance Metric (Clustering Evaluation)

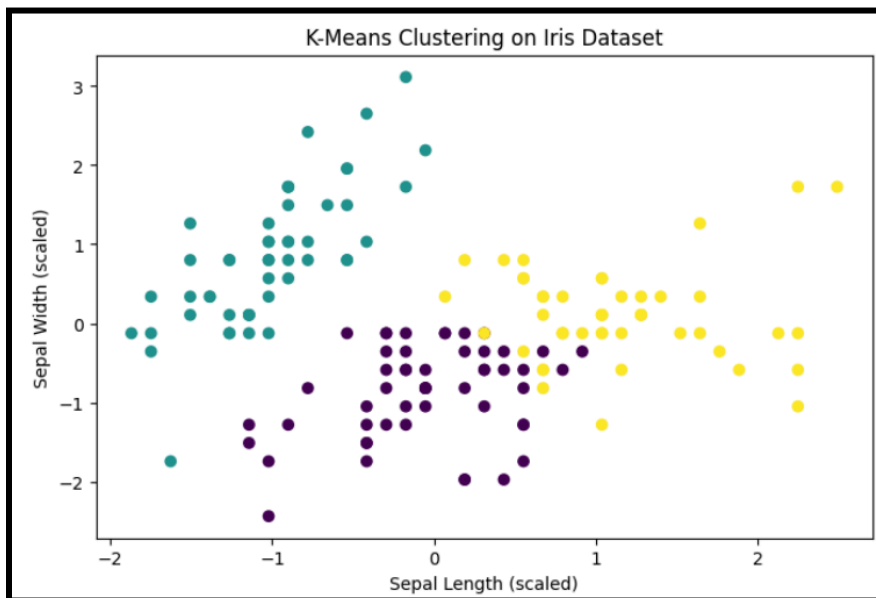
Since clustering is unsupervised, we use:

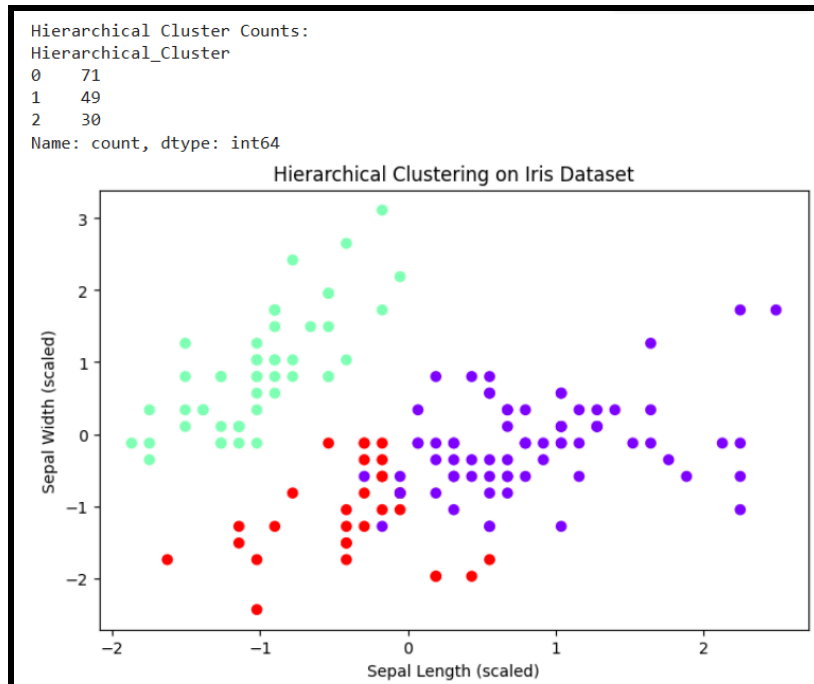
- **Inertia (WCSS)** → for K-Means
- **Visual Analysis** → cluster separation
- *(Optional)* Accuracy comparison with true labels

Example (approx values):

- Optimal clusters (K): **3**
- K-Means Inertia: ~78–100 (depends on scaling)

Output





Conclusion

Both K-Means and Hierarchical Clustering successfully formed meaningful clusters by grouping similar data points together based on feature similarity. The optimal number of clusters was found to be 3, which correctly matched the actual species categories in the dataset, indicating good clustering performance.

K-Means proved to be faster and more efficient, especially for larger datasets, due to its iterative centroid-based approach. On the other hand, Hierarchical Clustering provided better interpretability through the dendrogram, allowing visualization of how clusters are formed step by step.

Overall, clustering techniques are highly effective for identifying hidden patterns and structures in unlabeled data, making them useful for exploratory data analysis and segmentation tasks.